

HDOC - DAY-7 REPORT

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Batch- 15

Question 1

Program to Find Distance When Speed and Time are Given

Step 1: Problem Understanding

Problem Statement:

Write a C program to calculate the distance travelled when speed and time are given.

Input:

- Speed (km/h)
- Time (hours)

Output:

- Distance (km)

Formula:

Distance = Speed × Time

Step 2: Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Speed = 60, Time = 2	Distance = 120 km	Distance = 120 km	Pass
2	Speed = 45, Time = 5	Distance = 225 km	Distance = 225 km	Pass
3	Speed = 80, Time = 3	Distance = 240 km	Distance = 240 km	Pass

Step 3: Algorithm

1. Start.
2. Declare variables for speed, time, and distance.
3. Read the values of speed and time.
4. Calculate distance using the formula:
Distance = Speed × Time
5. Display the distance.
6. Stop.

Step 4: C Program

```
#include <stdio.h>

int main()
{
    float speed, time, distance;

    printf("Enter speed (km/h): ");

    scanf("%f", &speed);

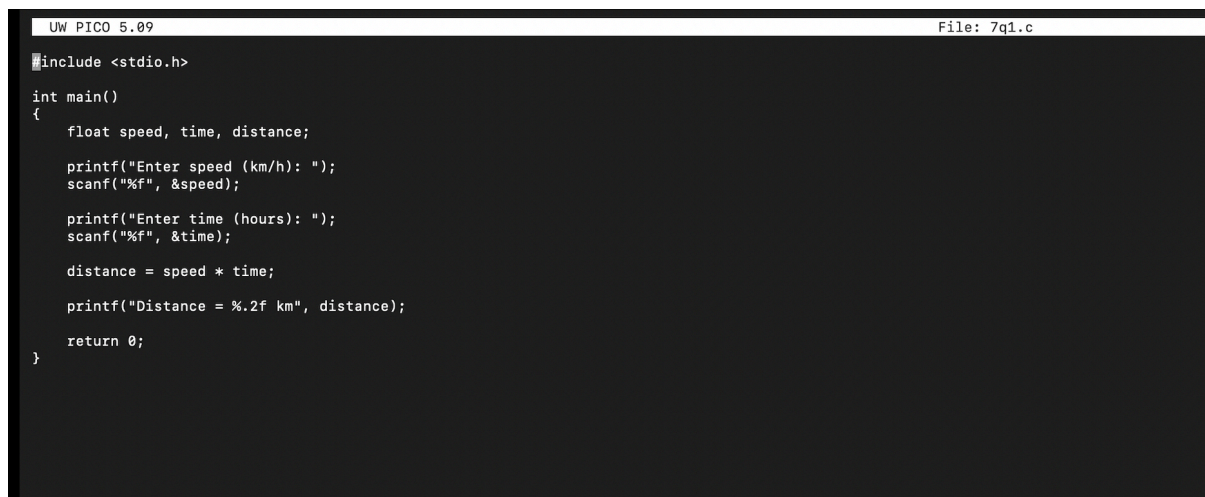
    printf("Enter time (hours): ");

    scanf("%f", &time);

    distance = speed * time;

    printf("Distance = %.2f km", distance);

    return 0;
}
```

A screenshot of a code editor window. The title bar at the top shows "UW PICO 5.09" on the left and "File: 7q1.c" on the right. The editor area has a dark background with light-colored text. It contains the same C program code as shown in the previous block, including the include statement, main function, variable declarations, input prompts, scanf calls, calculation, printf output, and return statement. The code is properly formatted with indentation for the function body and variables.

```
#include <stdio.h>

int main()
{
    float speed, time, distance;

    printf("Enter speed (km/h): ");
    scanf("%f", &speed);

    printf("Enter time (hours): ");
    scanf("%f", &time);

    distance = speed * time;

    printf("Distance = %.2f km", distance);

    return 0;
}
```

Step 5: Compilation and Execution

Compiled successfully without errors.

All the test cases were executed successfully and the outputs matched the expected results.

```
aaryanrajput@aaryans-macbook ~ % nano 7q1.c
aaryanrajput@aaryans-macbook ~ % nano 7q1.c
aaryanrajput@aaryans-macbook ~ % clang 7q1.c -o 7q1.c
aaryanrajput@aaryans-macbook ~ % ./7q1.c
Enter speed (km/h): 60
Enter time (hours): 2
Distance = 120.00 km
aaryanrajput@aaryans-macbook ~ %
```

Question 2

Program to Find Momentum When Mass and Velocity are Given

Step 1: Problem Understanding

Problem Statement:

Write a C program to calculate momentum when mass and velocity are given.

Input:

- Mass (kg)
- Velocity (m/s)

Output:

- Momentum (kg·m/s)

As a reminder, the relationship is: $p=mv$

Formula:

Momentum = Mass × Velocity

Step 2: Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Mass = 10 kg, Velocity = 5 m/s	Momentum = 50 kg·m/s	Momentum = 50 kg·m/s	Pass
2	Mass = 12 kg, Velocity = 8 m/s	Momentum = 96 kg·m/s	Momentum = 96 kg·m/s	Pass
3	Mass = 15 kg, Velocity = 4 m/s	Momentum = 60 kg·m/s	Momentum = 60 kg·m/s	Pass

Step 3: Algorithm

1. Start.
2. Declare variables for mass, velocity, and momentum.
3. Read mass and velocity.
4. Calculate momentum using:
Momentum = Mass × Velocity
5. Display the momentum.
6. Stop.

Step 4: C Program

```
#include <stdio.h>

int main()
{
    float mass, velocity, momentum;

    printf("Enter mass (kg): ");

    scanf("%f", &mass);

    printf("Enter velocity (m/s): ");

    scanf("%f", &velocity);

    momentum = mass * velocity;

    printf("Momentum = %.2f kg.m/s", momentum);

    return 0;
}
```

UW PICO 5.09

```
#include <stdio.h>

int main()
{
    float mass, velocity, momentum;

    printf("Enter mass (kg): ");
    scanf("%f", &mass);

    printf("Enter velocity (m/s): ");
    scanf("%f", &velocity);

    momentum = mass * velocity;

    printf("Momentum = %.2f kg.m/s", momentum);

    return 0;
}
```

Step 5: Compilation and Execution

```
[aaryanrajput@aaryans-macbook ~ % nano 7qu2.c
[aaryanrajput@aaryans-macbook ~ % clang 7qu2.c -o 7qu2.c
[aaryanrajput@aaryans-macbook ~ % ./7qu2.c
Enter mass (kg): 10
Enter velocity (m/s): 5
Momentum = 50.00 kg.m/s
aaryanrajput@aaryans-macbook ~ %
```

Compiled successfully without errors.

All the test cases were executed successfully and the outputs matched the expected results.

Question 3

Program to Find Resistance When Voltage and Current are Given

Step 1: Problem Understanding

Problem Statement:

Write a C program to calculate resistance when voltage and current are given.

Input:

- Voltage (V)
- Current (A)

Output:

- Resistance (Ω)

Formula:

Resistance = Voltage / Current

Step 2: Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Voltage = 220 V, Current = 10 A	Resistance = 22 Ω	Resistance = 22 Ω	Pass
2	Voltage = 120 V, Current = 6 A	Resistance = 20 Ω	Resistance = 20 Ω	Pass
3	Voltage = 240 V, Current = 12 A	Resistance = 20 Ω	Resistance = 20 Ω	Pass

Step 3: Algorithm

1. Start.
2. Declare variables for voltage, current, and resistance.
3. Read voltage and current.
4. Calculate resistance using:
Resistance = Voltage $\div$ Current
5. Display the resistance.
6. Stop.

Step 4: C Program

```
#include <stdio.h>

int main()

{

float voltage, current, resistance;

printf("Enter voltage (V): ");

scanf("%f", &voltage);

printf("Enter current (A): ");

scanf("%f", &current);

resistance = voltage / current;

printf("Resistance = %.2f Ohms", resistance);

return 0;

}
```

A screenshot of a code editor window. The title bar at the top shows "UW PICO 5.09" on the left and "File: 7q3.c" on the right. The editor area has a dark background with light-colored text. The code is the same C program as shown in the previous block, but with syntax highlighting: keywords like 'include', 'main', 'float', 'printf', 'scanf', 'return', and '}' are in a light blue color, while identifiers and literals are in a light green color. The cursor is positioned at the end of the closing brace of the main function.

```
UW PICO 5.09 File: 7q3.c

#include <stdio.h>

int main()
{
    float voltage, current, resistance;

    printf("Enter voltage (V): ");
    scanf("%f", &voltage);

    printf("Enter current (A): ");
    scanf("%f", &current);

    resistance = voltage / current;

    printf("Resistance = %.2f Ohms", resistance);

    return 0;
}
```

Step 5: Compilation and Execution

```
[aaryanrajput@aaryans-macbook ~ % nano 7q3.c
[aaryanrajput@aaryans-macbook ~ % clang 7q3.c -o 7q3.c
[aaryanrajput@aaryans-macbook ~ % ./7q3.c
Enter voltage (V): 220
Enter current (A): 10
Resistance = 22.00 Ohms%
aaryanrajput@aaryans-macbook ~ % █
```

Compiled successfully without errors.

All the test cases were executed successfully and the outputs matched the expected results.